We could write a scheduled job to do this for us. But fortunately, Elastic recently released a new feature called *index lifecycle management* (ILM). As the name suggests, ILM is about managing indexes based on a lifecycle *policy*. The policy is a definition declaring rules that are executed by the engine when the conditions of the rules are met. For example, we can define rules based on rolling over the current index to a new index when

- The index reaches a certain size (40 GB, for example)
- The number of documents in the index reaches, say, 10,000
- The day rolls over

Before we start scripting the policy, let's examine the lifecycle of an index: the phases an index progresses through based on criteria and conditions.

6.9.1 Index lifecycle

An index has five lifecycle phases: hot, warm, cold, frozen, and delete, as shown in figure 6.11. Let's briefly describe each of these phases:

- *Hot*—The index is in full operational mode. It is available for reads and writes, thus enabling the index for both indexing and querying.
- *Warm*—The index is in read-only mode. Indexing is switched off, but the index is open for querying so it can serve search and aggregation queries.
- *Cold*—The index is in read-only mode. Similar to the warm phase, indexing is switched off but the index is open for querying, although queries are expected to be infrequent. When the index is in this phase, search queries may result in slow response times.

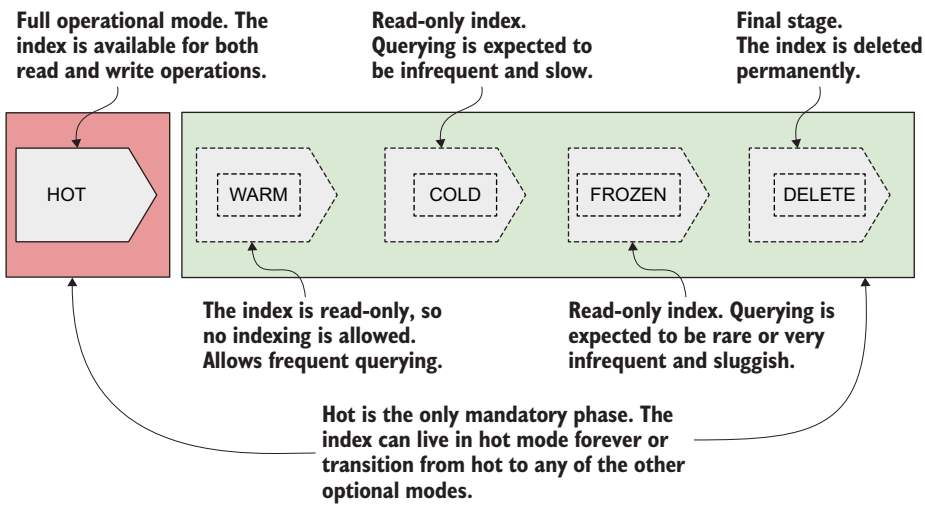


Figure 6.11 Lifecycle of an index